DESIGN EVALUATION 1

XR Whiteboard User Testing Guide and Round 1 Results

Name	Date	Prototype	Round
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What this evaluation tests

This evaluation tests whether users can find the wall whiteboard and table paper, choose the correct tools, draw a horizontal line, vertical line, and circle, move around the classroom, and understand the controls. It also checks drawing quality, table tool visibility, paper orientation, navigation, comfort, and confidence.

Testing aim and success measures

I want to learn whether users can complete the main classroom workflow without unnecessary help and whether the two drawing surfaces feel useful for study and teaching.

Metric	Success target
Find both surfaces	User reaches the wall whiteboard and a table paper.
Draw on both surfaces	User completes a horizontal line, vertical line, and circle on each surface.
Stroke quality	Strokes are continuous and smooth, with no dots, gaps, artefacts, or unwanted marks.
Tools	User changes colour, erases, and clears the intended board or paper.
Navigation	User can move to a table and understand the teleport or movement control.
Confidence	User reports confidence and explains any difficulty or improvement.

Test setup and task sequence

- Use the XR classroom scene with a blank wall whiteboard and blank table paper. Test with the same task order for each participant.
- Use a short moderated test with think aloud feedback. Do not lead the participant unless they are stuck or uncomfortable.
- P1, P2, and P3 consented to audio recording and have full transcript PDFs. P4 and P5 were recorded with checklists only.

Task	Instruction	Observe
1. Explore	Look around and identify the whiteboard and paper.	Does the user understand the classroom purpose?
2. Whiteboard	Draw a horizontal line, vertical line, and circle. Change colour, erase, and clear.	Stroke quality, tools, ray, and clear confirmation.
3. Table	Move to a table and explain what the paper is for.	Teleport visibility, view direction, and expectations.
4. Paper	Open TOOLS, choose Pencil, draw the three shapes, erase, and clear paper.	Tool visibility, paper orientation, and paper strokes.

Questions used

- What do you expect to do here, and what do you think this paper or button is for?
- What was the easiest part and what was the most difficult part?
- How did drawing on the paper compare with drawing on the whiteboard?
- Did you notice any glitches, gaps, unwanted marks, or controls that were hard to see?
- What is the most important improvement, and how confident are you from 1 to 5?

Round 1 results

Five classmates tested the prototype. The recorded sessions show detailed behaviour and comments. The checklist-only sessions show task completion and satisfaction but do not provide full quotes or confidence scores.

Participant	Evidence	Result	Main note
P1	Full transcript	Completed all core tasks. Confidence 4 out of 5.	Whiteboard was smooth and easiest. Paper felt slightly glitchy. Table tools and teleport were not immediately visible.
P2	Full transcript	Completed all core tasks. Confidence 5 out of 5.	Whiteboard was easier. Paper drawing felt smooth, but table tools were initially hard to notice.
P3	Full transcript	Completed all core tasks. Confidence 5 out of 5.	Understood the tools, but found looking up and down difficult.
P4	Checklist only	Completed all core tasks and was satisfied.	No full transcript or quote was created.
P5	Checklist only	Completed all core tasks and was satisfied.	No full transcript or quote was created.

Factual observations

- All five participants completed the planned core workflow on the wall whiteboard and table paper.
- The wall whiteboard was the easiest surface for the recorded participants. All participants could select tools, draw, erase, and clear.
- P1 reported slightly glitchy paper drawing. P2 described the paper drawing as smooth. This difference needs another focused retest.
- P1 and P2 needed to move closer or look more carefully before seeing the table tools. P3 also needed to inspect the table tools before using them.
- P1 did not notice the teleport control at first. P2 and P3 found navigation more clearly.
- Participants described the idea as useful for students who lose focus during online classes and for homeschooling because it can feel like a classroom.

Analysis and insights

Theme	Evidence	Meaning for the XR whiteboard
Useful learning context	Participants connected the classroom, whiteboard, and paper to studying, online learning, focus, and homeschooling.	Keep the classroom feeling and two-surface workflow.
Whiteboard is clearer	The whiteboard was easiest and felt smooth for P1, P2, and P3.	Keep the wall board as the main shared teaching surface.
Table tools lack visibility	P1, P2, and P3 needed closer inspection or movement before using the table tools.	Make the menu larger, clearer, and easier to read from the seated view.
Paper drawing needs retest	P1 noticed a glitch while P2 reported smooth drawing.	Repeat the paper test and compare controller, hand, editor, and headset input.
Navigation and view need support	Teleport was missed by P1. P3 found looking up and down difficult.	Improve teleport hints and add a clearer vertical view interaction.

Check the testing aims

Aim	Status	Evidence
Users can find and use both surfaces.	Validated with issue	All five completed the workflow, but table tool visibility was inconsistent.
Users can draw three shapes on both surfaces.	Validated with issue	All five completed the tasks. P1 reported slightly glitchy paper drawing.
Users understand tools and clear actions.	Validated with issue	All five completed the controls, but the paper tools were hard to notice.
Navigation and UI support confident use.	Partly validated	Confidence was high, but teleport visibility and vertical view need improvement.

Concept iteration

Insight from testing	Design response	Reason
Table tools are hard to read or find.	Make the TOOLS button and paper menu larger, higher contrast, and visible from the normal seated view. Add a short hint.	Users completed the task after moving or receiving help, so the function works but the affordance needs improvement.
Paper drawing was reported as slightly glitchy by P1.	Retest paper input, inspect the exact stroke failure, and compare it with the internal continuous-stroke validation.	Different participants gave different paper feedback, so the cause is not fully confirmed.
Teleport was missed by P1.	Enlarge the navigation control and add a visible first-use hint.	A participant did not know movement was available until the control was pointed out.
Looking up and down was difficult for P3.	Add or test a clearer vertical view control and check the seated camera angle.	The participant could use the classroom but found the view range uncomfortable.
The concept supports focused study.	Keep the classroom, shared whiteboard, and individual paper notes. Explore more tables and multiplayer later.	Participants linked the concept to online learning, homeschooling, and staying focused.

Updated concept description

The XR Study Whiteboard provides a classroom style study space where users can follow a shared lesson on the wall whiteboard while writing personal notes on paper at a table. The first prototype shows that the core workflow is understandable and useful, especially for students who are easily distracted during online classes or who study at home. The next version should improve table tool visibility, paper drawing reliability, teleport discovery, and vertical viewing before adding larger classroom or multiplayer features.

Reflection and next steps

- The same task order made the five sessions easy to compare. Think aloud feedback revealed problems that a task success score alone would miss.
- The facilitator sometimes gave hints about moving closer or finding tools. In the next round, use neutral prompts first and record every hint as assistance.
- Run another focused paper drawing test with horizontal lines, vertical lines, and circles. Record whether the issue is visible and whether it depends on input method or device.
- Improve the table menu size and contrast, add a teleport hint, and test a comfortable way to look up and down.
- Test the revised build with more participants and a real Quest headset. Continue to keep P4 and P5 checklist records separate from full transcripts.

Evidence record and final checklist

Evidence	Status
P1, P2, and P3 full transcript PDFs	Stored in Design Evaluation/Interview Transcripts/
P4 and P5 checklist-only notes	Stored in Participant04-05_Checklist.md
Core task results, insights, and design changes	Included in this guide
Next prototype and next test	Clear: improve table tools, paper drawing, navigation, and view control

The evidence supports the IP1 aim, but the paper drawing issue and table tool visibility should be tested again after the next changes. This report does not invent transcripts or quotes for the two checklist-only sessions.